

Gradient ascent for logistic regression

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1 Likelyhood function

$$\ell(w) = \prod_{i=0}^{n-1} P(y_i|x_i, w),$$

Derivative of likelyhood function (or rather the log of the likelyhood function)

$$\frac{\partial \ell(w)}{\partial w_j} = \sum_{i=0}^{n-1} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w))$$